

- 5 Fig. 5.1 shows the electric field in the region between two points P and Q. The electric potential at P and at Q are +400 V and -400 V respectively.

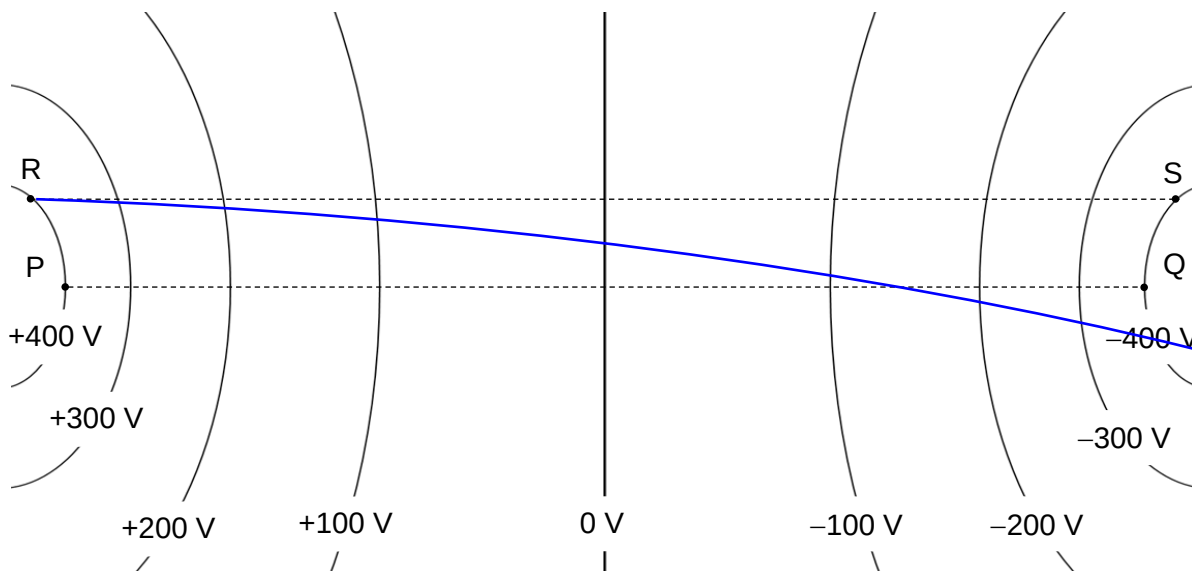


Fig. 5.1

- (a) Define *electric potential* at a point.

Electric potential at a point is the work done by external agent per unit positive charge on a small test charge to bring it from infinity to that point, (without change in its kinetic energy.)

[2]

- (b) Describe how the direction and magnitude of the electric field strength varies along the line PQ.

From P to Q, the direction of the field is always towards Q, and the magnitude will decrease to a minimum and then increase.

[2]

- (c) An electron is projected from P towards Q with a speed of $2.2 \times 10^7 \text{ m s}^{-1}$. Calculate its speed when it reaches Q.

Loss in KE = Gain in EPE

$$\frac{1}{2} m_e (u^2 - v^2) = q \Delta V \rightarrow \frac{1}{2} \times 9.1 \times 10^{-31} \times ((2.2 \times 10^7)^2 - v^2) = 1.6 \times 10^{-19} \times 800$$

$$v = 1.4 \times 10^7 \text{ m s}^{-1}$$

speed = m s^{-1} [3]

- (d) Another electron is projected from R towards S. Explain why this electron will not move in the path RS.

The direction of the electric field is not along the line RS, thus the electron will accelerate in a direction different from its velocity and move in a curved path. [1]

- (e) Sketch a possible path taken by the electron in (d). [1]

[Total: 9]

- 6 (a) A cell of e.m.f. 1.5 V and internal resistance $0.25\ \Omega$ is connected in series with a resistor R